

API Documentation

REST & WebSocket API reference. Interactive docs: api.169-58-221-49.nip.io/docs (OpenAPI 3.1 Swagger UI)

1. Conventions

- Base URL: <https://api.169-58-221-49.nip.io>
- Auth: all endpoints except / and /health require Authorization: Bearer <JWT> (Keycloak realm soc-platform). /internal/verify-api-key uses X-API-Key instead.
- Response shape: most endpoints fail soft — internal errors return 200 OK with {"error": "..."} rather than a 5xx; clients should always check for an error key.
- Rate limiting: Traefik applies 50 req/avg, 100 burst, per minute at the edge.

2. Endpoints

System

Method / Path	Description	Auth
GET /, /health, /metrics	Root, health check, Prometheus metrics	None

Threat Intelligence

Method / Path	Description	Auth
GET /threatintel/ip/{ip}, /abuseipdb/{ip}, /otx/{ip}, /urlscan/{domain}, /shodan/{ip}	Raw per-provider lookups	JWT
GET /threatintel/check/{ip}	Unified check — fans out to VT/AbuseIPDB/OTX/Shodan, computes overall_verdict + malicious_signals, caches in Redis, indexes in Elasticsearch, auto-creates a TheHive case, broadcasts over WebSocket	JWT
GET /threat-intel/history	Last 20 unified check results	JWT

AI Features

Method / Path	Description	Auth
GET /ai/explain, /executive-summary, /mitre-map, /cve/{cve_id}, /recommend, /threat-report, /rag-explain — all {ip}-scoped, ?provider=	Explanation, ≤5-sentence exec summary, MITRE mapping, CVE explain, action list, structured report, RAG-augmented explain	JWT
POST /ai/malware-explain	Body: {hash?, filename?, url?} — explains a possible malware artifact + next steps	JWT

Autonomous Agents (LangGraph)

Method / Path	Description	Auth
GET /agents/threathunt/{ip}	Threat Hunting Agent — VirusTotal + MISP, investigation report	JWT
GET /agents/triage/{ip}	Incident Triage Agent — classifies Critical/High/Medium/Low, auto-assigns	JWT
POST /agents/malware-investigate	Body: {hash?, filename?, url?} — risk level + containment recommendation	JWT
GET /agents/exec-report/{period}	Executive Reporting Agent — weekly/monthly/quarterly, stores report in MinIO	JWT

Reports, Detection Rules, Monitoring

Method / Path	Description	Auth
GET /reports/{filename}/download-url	MinIO presigned download URL	JWT
POST /yara/scan, POST /sigma/evaluate, GET /sigma/rules	Scan text against YARA rules; evaluate a log event against Sigma rules; list loaded Sigma rules	JWT
POST .../falco/ingest, .../suricata/ingest, .../zeek/ingest	Manually trigger ingestion from falco.log / eve.json / notice.log	JWT
GET .../falco/events, .../suricata/alerts, .../zeek/notices	Last 50 stored events per source	JWT
GET /security-monitoring/wazuh/status, /wazuh/agents, /status	Wazuh health, enrolled agents, aggregate health of all monitoring tools (polled every 15s by the Network Status page)	JWT

Alerts, Cases, Threat Map, Search

Method / Path	Description	Auth
GET /alerts/unified	Merges threat-intel + Suricata + Zeek + Falco + system alerts, newest-first, top 50	JWT
WS /ws/alerts?token=	Live WebSocket stream of new_alert events	JWT (query param)
GET /cases, /cases/{case_id}	List / detail of TheHive cases	JWT
GET /threat-map	Country-aggregated verdict counts + raw points, last 200 records	JWT
GET /search?q=	Unified full-text search across Elasticsearch indices	JWT

Assets & Organizations

Method / Path	Description	Auth
GET/POST /assets, DELETE /assets/{id}	List / create / delete assets	JWT
GET/POST /organizations, DELETE /organizations/{id}	List / create / delete tenants	JWT

Internal

Method / Path	Description	Auth
GET /internal/verify-api-key	Validates an X-API-Key against API_KEYS env var — separate from the Keycloak JWT path	API Key

3. Key Schemas

Schema	Fields
AssetCreate	name (required), ip_address, asset_type (default "server"), owner, criticality (default "medium")
OrganizationCreate	name (required), description
MalwareExplainRequest	hash, filename, url — all optional, at least one expected
SigmaEvaluateRequest	log_event: object
YaraScanRequest	content: string
Unified threat check response	ip, sources_checked[], overall_verdict (clean/suspicious/malicious), malicious_signals, details{}, thehive_case? (present only if created)

4. Interactive Documentation

The backend auto-generates a full OpenAPI 3.1 spec at /docs (Swagger UI, with an Authorize button for a bearer token) and the raw spec at /openapi.json. This is the authoritative reference for exact request/response schemas. The tables above summarize it for offline reading.